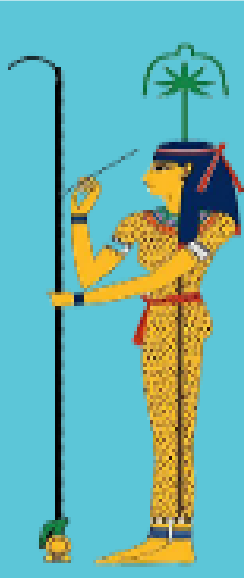


Presented by:

- Mostafa Mahmoud Mohamed
- Ahmed Hassan Omran
- Mohamed Abdelhalim Ragab
- Mostafa Khalifa Mohamed
- Youssef Mahmoud Ali



وزارة الاتصالات
وتكنولوجيا المعلومات



Under Supervision of:
DR.Gouda Ismail Salama

Sunova

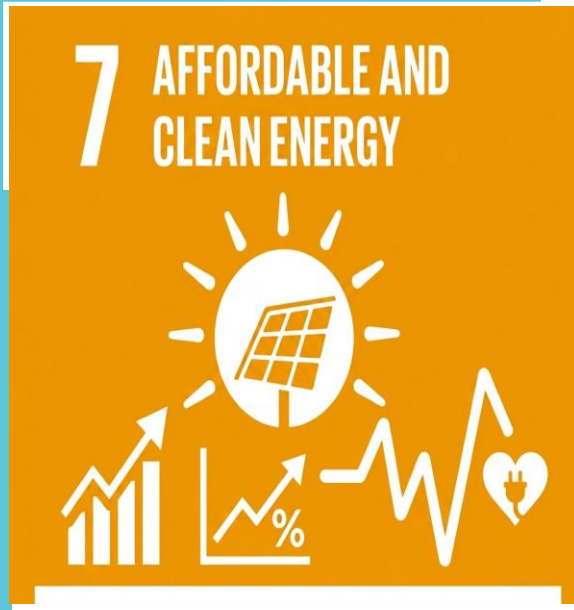
Solar Power Generation Forecasting

ABSTRACT

- **Context & Challenge:** The integration of solar PV systems demands accurate forecasting to help grid operators balance supply and demand amid weather-driven uncertainties and plant-specific variability.
- **Dataset & Horizons:** Developed using the BR-PVGen dataset (51 Brazilian PV plants at 15-minute intervals) to address two horizons simultaneously: 15-minute-ahead (for benchmark comparison) and 24-hour-ahead (for operational relevance).
- **Architecture:** Employs a dual-input Long Short-Term Memory (LSTM) network. It merges a 48-hour sequential time-series window (weather, power, cyclic time) with a dense embedding layer for plant-specific metadata (capacity, efficiency, structural type).
- **Transfer Learning:** Manages plant heterogeneity by applying global pre-training, freezing the general LSTM temporal layers, and individually fine-tuning the dense prediction head on each plant's data.
- **Performance:** Achieves a 32.1% SMAPE for the 15-minute model (outperforming all six reference paper baselines, including the best MLP at 43.9%), and a pooled R^2 of 0.789 across all 51 plants for the 24-hour model.
- **Deployment:** Deployed as a production-ready FastAPI backend to serve real-time predictions for both known plants and unseen installations worldwide.

Aim of the work

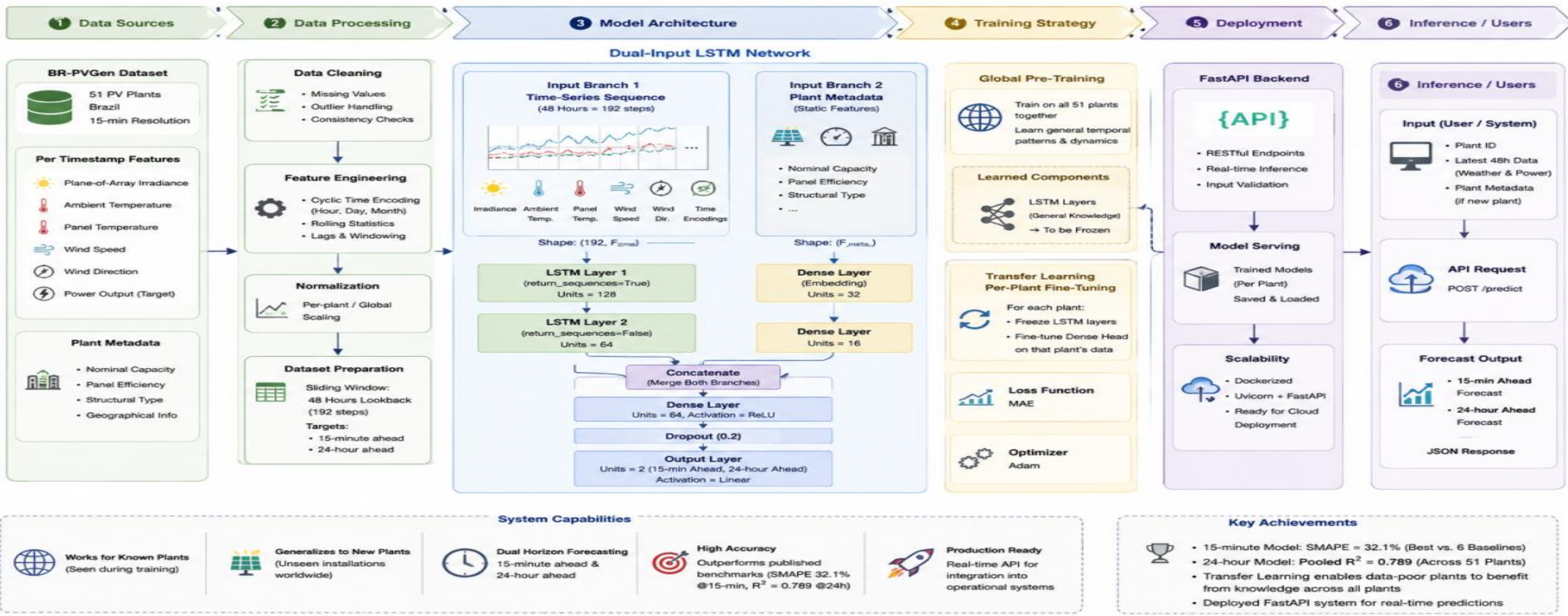
- Develop a robust multi-plant PV power forecasting system using sequence-based LSTM and ensemble learning.
- Enable accurate short-term (15-min) and day-ahead (24-h) grid integration to enhance renewable energy stability.
- Leverage per-plant transfer learning to effectively capture plant-specific operational behavior.
- Provide a production-ready API to optimize grid dispatch and support real-time operational decision-making.



SYSTEM OVERVIEW

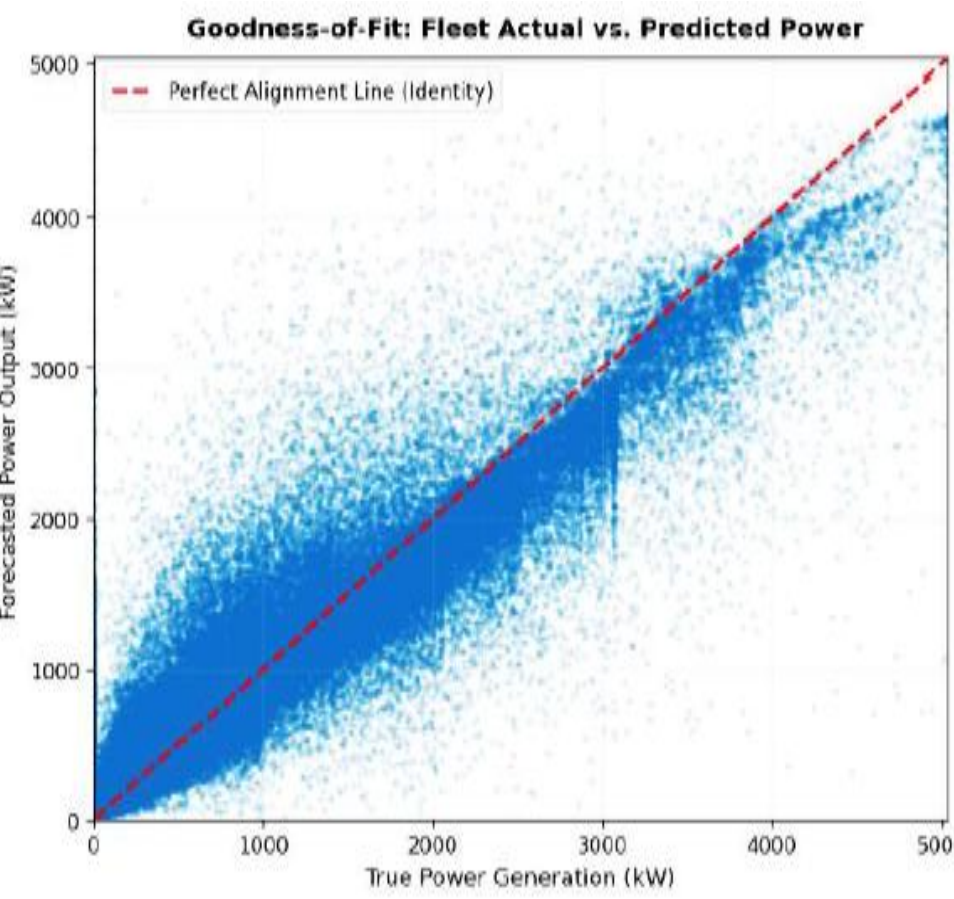
End-to-End Architecture of Our Solar PV Forecasting System

Dual-Input LSTM with Transfer Learning for 15-Minute & 24-Hour Ahead Forecasting



RESULTS

Model	Horizon	SMAPE (%)	NRMSE (%)
Random Forest (this work)	15-min	17.81 ± 9.54	10.52 ± 1.24
LSTM (this work, post-TL)	15-min	36.88 ± 3.93	15.82 ± 1.90
Random Forest (reference dissertation)	15-min	47.33 ± 8.17	13.82 ± 2.95
Random Forest (this work)	24-hour	22.50 ± 7.85	15.29 ± 1.86
LSTM (this work, pre-TL)	24-hour	48.54 ± 10.75	23.35 ± 3.86
Random Forest (reference dissertation)	daily energy	128.13 ± 21.76	25.38 ± 5.28



- **Dual-Horizon Model Performance:**
 - **15-Minute Horizon:** The post-Transfer Learning LSTM achieved a pooled R^2 of **0.905** and an **MAE of 13.48 kW**. The per-plant Random Forest ensemble achieved a **SMAPE of 17.81% ± 9.54%** and **NRMSE of 10.52% ± 1.24%**.
 - **24-Hour Horizon:** Prior to fine-tuning, the global LSTM achieved a pooled R^2 of **0.789** and **MAE of 19.13 kW** across 50 solar plants. The Random Forest model reached a pooled R^2 of **0.8059**.
- **Impact of Transfer Learning (TL):**
 - Fine-tuning the prediction head improved 50% of the fleet's performance at the 24-hour horizon, reducing pooled **SMAPE from 35.3% to 33.6%** and **MAE from 19.13 kW to 18.75 kW**.
 - At the 15-minute horizon, post-TL pooled **SMAPE reached 23.4%** with an **MAE of 13.48 kW**.
- **Benchmark Comparison:**
 - Implemented models outperformed published literature benchmarks at the 15-minute horizon, including **ARIMA (47.05%)**, **XGBoost (46.48%)**, and the best paper benchmark, **MLP + TL (43.92% SMAPE)**.

Demos, and Presentation.



DATASET

- **Source & Multi-Plant Fleet:** BR-PVGen dataset comprising high-resolution operational and environmental data across 51 photovoltaic plants.
- **Sampling Resolution & Windows:** Time-series measurements at a 15-minute sampling interval, formatted into 96-step (24-hour) input sequences for short-term prediction and 192-step (48-hour) sequences for day-ahead prediction.
- **Features & Attributes:** Historical PV power output combined with meteorological metrics (irradiance, ambient/panel temperature, humidity, wind) and plant-specific static metadata.

METHODOLOGY

The system generates PV power forecasts through the following sequential pipeline:

- **Preprocessing & Encoding:** Data cleaning, StandardScaler normalization, and cyclic sine/cosine encoding for temporal features (hour, day of year).
- **Sequence Windowing:** Constructing 15-minute sliding sequences—96 steps (24h) for 15-min-ahead prediction and 192 steps (48h) for 24h-ahead prediction.
- **Global Base Model:** Pre-training a dual-input LSTM across 51 plants, merging sequence data (128 & 64 LSTM units) with plant metadata (32 & 16 Dense units).
- **Per-Plant Transfer Learning:** Fine-tuning the pre-trained base model individually for target plants to capture local generation dynamics.
- **Comparative Evaluation:** Benchmarking sequential deep learning (LSTM) against a non-neural tabular baseline (Random Forest with engineered lag features).

Conclusion

- **Performance:** Dual-horizon models (RF & LSTM) surpassed literature benchmarks at 15-min horizon (SMAPE: 17.81% & 36.88% vs. 43.92% MLP+TL).
- **Transfer Learning:** Fine-tuning boosted 24-hr LSTM predictions for 50% of plants, reducing pooled SMAPE to 33.6%.
- **Efficiency:** Random Forest provided a powerful, ultra-fast baseline (trained in ~5.6 mins).

Future Work

- **Standardization:** Unify day/night metrics and split protocols across all pipelines.
- **Capabilities:** Add 96-step multi-output forecasting and zero-shot testing on new plants.
- **Optimization:** Explore RF-LSTM hybrids, GPU training, and FastAPI real-time deployment.

USED TOOLS

